

Part II

Enumerative Resonance, Energy as Residue, and Physical Calibration

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11 September 2026

A standalone continuation of *Lambda, Reference, and the Spatiotemporal Descriptive Frame*.

This paper develops the arithmetic, spectral, and physical applications in their own right. It integrates the corrected enumerative-resonance and energy-as-residue notes, with the infinite-registry framework supplying the proof and representation discipline.

Verification included in this document

23 Lean theorem declarations checked in Lean 4.33.0 with Mathlib; complete source, theorem checks, and axiom reports. Sixteen fresh computational check families, with exact finite-domain tests and numerical model receipts. Reproduction files and the three source notes accompany the PDF as embedded attachments. Verification applies to the stated mathematical claims and declared

model calculations. No new observational or laboratory dataset accompanies this edition. No inherited document hash serves as a receipt.

Abstract

The descriptive-frame program gains physical content when a representation comes with an explicit target, a transformation law, an operator, and a measurement calibration. This paper develops that chain. In a cyclic frame, the orbit period of a step equals the modulus divided by the corresponding greatest common divisor. Primality then admits equivalent descriptions through full nonzero orbits, the absence of intermediate divisors, and the absence of proper non-trivial subgroups. Character theory carries this enumeration into phase structure; an operator carries phase structure into a spectrum.

For the cyclic Laplacian, the operator identifies a phase with its inverse. Consequently two frames of moduli n and m , with $g = \gcd(n, m)$, share g oriented phases, $\lfloor g/2 \rfloor + 1$ distinct spectral values, and $2g - \gcd(g, 2)$ ordered mode coincidences. These three counts answer different questions. Their separation supplies a concrete example of recoverable structure and irreversible descriptive loss. Calibrated energy levels, relative quantum phase, and perturbative splitting then give physical tests of the operator model.

The same discipline applies to spacetime. Lorentz transformations preserve the joint interval while mixing spatial and temporal coordinates; neither coordinate alone determines the causal classification. SI calibration couples length and time through the fixed numerical value of c , while energy calibration supplies the dimensional information absent from a bare spectrum. Finite propagation, scale selection, and finite-band relaxation further delimit the physical claims. The resulting “master key” functions as a repeatable method for proving what survives a change of description and identifying what extra information a target requires. It does not replace a physical model’s dynamics or its measurement evidence.

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1 Standing of Part II and the route from structure to physics

The published Part I supplies the logical framework for reference and descriptive dependence. Part II supplies a sustained development of the prime, resonance, energy, and spacetime applications. Its immediate source base comprises *The Enumerative Resonance Correspondence*, V2; *Energy as Residue – VUPDATED*, the updated edition; and *The Infinite Registry and Indexed-Residue Proof Engine*, Lab Note V2 [2, 3, 4]. The earlier Masterkey draft contributes an architectural aim: one disciplined method should work across domains. Its superseded assertions do not serve as premises here.

The central route has five stages:

1. Declare the states, relations, and admissible re-descriptions.
2. Prove the invariant or the information-loss obstruction.
3. Select an operator or dynamical law that turns structure into a prediction.
4. Supply the dimensional calibration and the actual readout.
5. Compare the resulting prediction with a stated computation or measurement.

Each stage adds information. A theorem about characters does not choose a Hamiltonian; a Hamiltonian in dimensionless units does not determine a joule; an exact model calculation does not supply an experimental dataset. Keeping those interfaces explicit strengthens the physical claim: each interface can itself carry a proof, a calibration procedure, or a falsifiable test.

Claim layer	Evidence supplied here	Extent of the claim
Arithmetic and representation	Written proofs; selected Lean theorems; exact finite checks	Universal statements where proved, finite coverage where enumerated
Spectral and dynamical models	Algebraic derivations; numerical linear algebra and perturbation tests	The declared operators, coupling, and autonomous evolution
Physical realization	SI definitions; Lorentz model; calibration and readout conditions	Conditional physical predictions and standard physical identifications
Empirical validation	No new observational dataset supplied	Historical reports remain attributed reports; fresh runs test mathematical models

The audit in Section 14 maps the exact formal coverage. A written theorem may extend beyond the included Lean encoding; the map records that difference. All numerical results printed as fresh checks come from the accompanying program and receipt.

2 The registry, the target, and the meaning of residue

2.1 Recoverability gives the first test

Let X carry the descriptions under consideration, let $P : X \rightarrow Y$ retain a chosen readout, and let $Q : X \rightarrow Z$ denote the target. Recoverability from P means that some decoder $f : Y \rightarrow Z$ satisfies

$$Q = f \circ P. \quad (1)$$

Proposition 2.1 (Collision obstruction). *If $P(x_1) = P(x_2)$ but $Q(x_1) \neq Q(x_2)$, no decoder in (1) exists.*

Proof. A decoder would give $Q(x_1) = f(P(x_1)) = f(P(x_2)) = Q(x_2)$, contradicting the hypothesis. $\square$

This elementary obstruction becomes physically substantive when P and Q name actual quantities. Below, P can mean a Laplacian spectral value and Q an oriented phase; alternatively, P can mean one spacetime coordinate and Q the boosted coordinate or causal class. The instances provide concrete witnesses, not a verbal analogy.

If a symmetry σ preserves every supplied predicate but changes the target, the same argument applies. Any construction from that unchanged data must also respect its symmetry. Direction, orientation, an origin, or an equality predicate may therefore require explicit supplementary data. Such supplements form part of the declared representation vocabulary.

2.2 An indexed cost, not an unqualified substance

For a target ϕ , let $\mathcal{R}(\phi)$ collect admitted representations and let $O(r) \in [0, \infty]$ measure a declared cost. Define

$$I_{\mathcal{R},O}(\phi) = \inf\{O(r) : r \in \mathcal{R}(\phi)\}, \quad \inf \emptyset = \infty. \quad (2)$$

If $\mathcal{R}_0(\phi) \subseteq \mathcal{R}_1(\phi)$ and the same cost applies to both, then

$$I_{\mathcal{R}_1,O}(\phi) \leq I_{\mathcal{R}_0,O}(\phi). \quad (3)$$

Every lower bound for the larger cost set also bounds the smaller one; this proves the inequality. The included Lean declaration formalizes it for extended nonnegative real costs.

An invariant under a re-description and an infimum of a cost play related but distinct roles. The first records what survives an equivalence; the second records how cheaply an admitted description can achieve a target. “Energy as residue” acquires precise content only after specifying which role applies, which operator measures energy, and which transformations count as admissible.

Positive cost for every member need not imply a positive infimum. For example, every $1/j$ for $j \geq 1$ exceeds zero, while their infimum equals zero. A uniform lower bound needs a separate argument. The Lean module includes the corresponding general counterexample using the positive real numbers.

2.3 A conservative proof engine

Fix a formal theory and a primitive checker. An expanded representation language remains conservative when every admitted new proof expands to an old proof with the same conclusion, and every old proof remains admitted. The theorem sets then coincide: inclusion gives one direction, expansion gives the other. The Lean module checks this implication without additional axioms.

A proof registry may lower represented cost through reusable lemmas, macros, or compressed descriptions. It does not thereby prove a previously unprovable statement of the fixed theory. Expanded primitive cost and compressed displayed cost require separate accounting. A running best cost b_N supplies a decreasing sequence of upper bounds; failure to find a proof by stage N supplies no unprovability certificate.

Finite exhaustive search also needs genuinely finite sublevel sets. A bound on inference count alone need not bound formula length. A finite alphabet together with a bound on total encoded length does give a finite search domain. If axioms have only an effective enumeration, the certificate must include sufficient enumeration evidence for the primitive checker. These conditions preserve the registry note’s intended audit discipline without assuming that its proposed full engine has already been implemented [4].

Verification connection. E13 checks residue monotonicity; E14 checks conservative expansion; E15 checks the failure of a uniform positivity inference. Part I supplies the general factorization framework; E09 and P01–P06 below instantiate it in spectra and spacetime.

3 Primes: inward divisibility and outward traversal

3.1 Exact orbit length

Take $n \geq 1$ and the additive cyclic group $C_n = \mathbb{Z}/n\mathbb{Z}$. A step k acts by $x \mapsto x + k$. Starting at zero produces $0, k, 2k, \dots$ modulo n .

Theorem 3.1 (Orbit period). *The least positive return index of step k equals*

$$p_n(k) = \frac{n}{\gcd(n, k)}. \quad (4)$$

The step visits every state exactly when $\gcd(n, k) = 1$.

Proof. Write $g = \gcd(n, k)$, $n = gn'$ and $k = gk'$, with $\gcd(n', k') = 1$. Then $jk \equiv 0 \pmod{n}$ exactly when $n' \mid jk'$, equivalently $n' \mid j$. The least positive such j equals n' . The distinct states before that return form the generated subgroup, so full coverage occurs exactly when $n' = n$. $\square$

The identity step $k = 0$ has period one. This exception matters whenever a statement refers to “every step.” For $n = 1$, the group contains only the identity; primality statements below therefore begin at $n = 2$.

Theorem 3.2 (Equivalent prime tests). *For $n \geq 2$, the following conditions are equivalent:*

1. n is prime.
2. No divisor d satisfies $2 \leq d < n$.
3. Every nonzero step in C_n traverses all n states.
4. C_n has no proper nontrivial subgroup.
5. C_n has no quotient whose order lies strictly between 1 and n .

Proof. The divisor condition restates primality. For prime n , every $0 < k < n$ is coprime to n , so Theorem 3.1 gives a full orbit. Conversely, a proper divisor d yields a nonzero step of period $n/d < n$.

For the subgroup statement, the inverse image of any subgroup under $\mathbb{Z} \rightarrow C_n$ has the form $d\mathbb{Z}$ with $d \mid n$. Its image equals $\langle d \rangle$ and has order n/d . Thus intermediate divisors and proper nontrivial subgroups correspond. The quotient by that subgroup has order d , proving the final equivalence. $\square$

The subgroup lattice needs a precise indexing convention. The map $d \mapsto \langle d \rangle$ reverses divisibility: $d_1 \mid d_2$ implies $\langle d_2 \rangle \subseteq \langle d_1 \rangle$. Indexing instead by subgroup order gives the order-preserving map $d \mapsto \langle n/d \rangle$.

These equivalences express the inward/outward prime correspondence. Inward factor search finds no intermediate divisor; outward traversal finds no early return from a nonzero step. A composite modulus still admits generators: $k = 1$ always generates, and every coprime step does so. What distinguishes a prime modulus is the universal quantifier over its nonzero steps.

3.2 What a change of number base preserves

Let $e : X \rightarrow Y$ be a bijective encoding and $T : X \rightarrow X$ the step operation. Transport the operation as $T_e = e \circ T \circ e^{-1}$. Induction gives

$$T_e^j(e(x)) = e(T^j(x)). \quad (5)$$

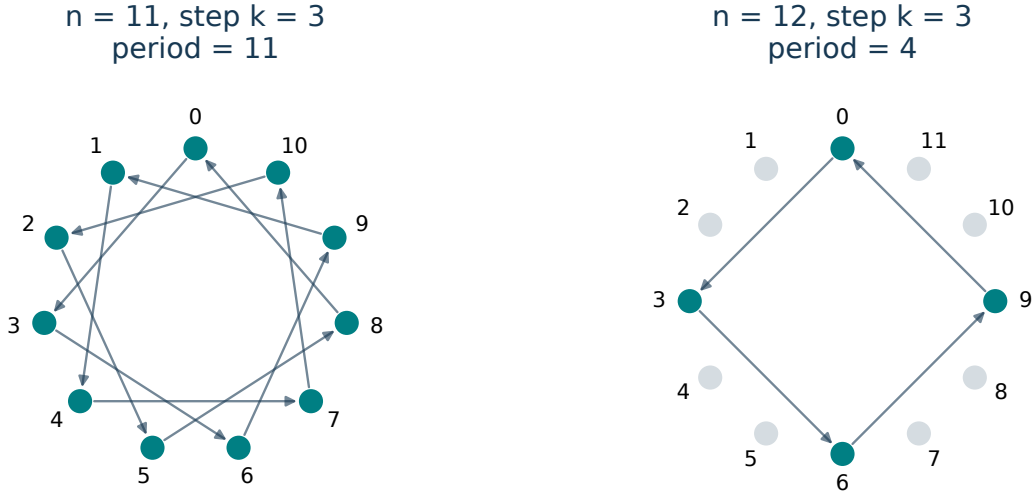


Figure 1: The same step $k = 3$ traverses all eleven states modulo 11, but only four states modulo 12. For 12, the orbit $0, 3, 6, 9$ witnesses the proper subgroup and the period $12/\gcd(12, 3) = 4$.

Injectivity then gives $T_e^j(e(x)) = e(x)$ if and only if $T^j(x) = x$. Return indices and minimal periods survive intact. Writing an integer in base two, ten, twelve, or sixty changes its numeral, while transported arithmetic preserves its divisors and primality.

Changing the modulus changes the group itself. Dropping digits or identifying states without preserving the necessary operations can also change the target. Thus the prime theorem concerns the subgroup and quotient structure of a specified cyclic group. The phrase “no faithful coarser frame” needs this qualification: an injective encoding into fewer states fails for every finite set, whereas the absence of an intermediate nontrivial cyclic quotient characterizes prime order.

This supplies a theorem-level prime result supported by the updated source [2]. It does not assert a prime-density formula, a new asymptotic distribution law, or a proof of the Riemann hypothesis. No such result follows from the orbit equivalences.

Verification connection. E01 formalizes the divisor characterization; E02 proves the prime coprimality step; E05–E06 prove conjugacy and return invariance. Fresh exact checks cover 19,900 orbits, all 198 candidate moduli from 2 through 199, subgroup orders for 99 moduli, and 7,076 transported-base cases. The complete cyclic subgroup and minimal-period arguments above remain written proofs.

4 Enumeration becomes resonance through characters

4.1 The Fourier dictionary

For $a, x \in C_n$, define $\chi_a(x) = e^{2\pi i ax/n}$. With the normalized inner product

$$\langle f, g \rangle = \frac{1}{n} \sum_{x=0}^{n-1} \overline{f(x)} g(x), \quad (6)$$

the characters form an orthonormal basis. Indeed, the inner product of χ_a and χ_b gives the normalized geometric sum with ratio $e^{2\pi i(b-a)/n}$. The ratio equals one when $a = b$ modulo n ; otherwise the sum vanishes. There are n orthonormal vectors in an n -dimensional space, proving completeness.

The character identity $\chi_a(x+y) = \chi_a(x)\chi_a(y)$ converts additive traversal into multiplicative phase. This exact correspondence gives the operative meaning of enumerative resonance: closure in the finite frame selects compatible phase modes. Standard finite-group character theory supplies the surrounding mathematical setting [5].

Characters of different groups have different domains. To compare their phases, explicitly embed their values in the common unit circle

$$\mu_n = \{z \in \mathbb{C} : z^n = 1\}. \quad (7)$$

Without that common carrier, saying that two characters “are the same” omits part of the comparison.

Theorem 4.1 (Shared phases and joint return). *For positive n, m and $g = \gcd(n, m)$,*

$$\mu_n \cap \mu_m = \mu_g, \quad |\mu_n \cap \mu_m| = g. \quad (8)$$

Two unit-step counters modulo n and m first return together at $\text{lcm}(n, m)$.

Proof. If $z^g = 1$, divisibility of n, m by g gives both root conditions. Conversely, Bézout coefficients $a, b \in \mathbb{Z}$ with $g = an + bm$ give $z^g = (z^n)^a(z^m)^b = 1$ for any common root. All roots are nonzero, so negative powers present no difficulty. The g roots $e^{2\pi i j/g}$ are distinct. For the counters, a common return requires both $n \mid t$ and $m \mid t$; the least positive such t equals the least common multiple. $\square$

The gcd and lcm answer dual-looking but different questions: shared compatible phase content versus the first common return index. If the counters carry non-unit steps, their individual periods become $p_n(k)$ and $p_m(\ell)$; the simultaneous return then uses the lcm of those periods.

If physical phase accumulates at frequencies ν_1 and ν_2 , a finite common return additionally requires a rational frequency ratio. For $\nu_1 = p\nu_0$ and $\nu_2 = q\nu_0$ with coprime positive integers p, q , the first joint full-cycle return occurs at $1/\nu_0$. Musical or mechanical relevance requires a declared frequency mapping and coupling; a large gcd alone does not determine perceived consonance or resonance amplitude.

4.2 The registry geometry leaves the operator open

The updated energy note places finite cyclic phase sets inside the circle action used in the Hopf construction, with the inclusions leading toward $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$ [3]. That geometry provides a common phase organization. It does not select a finite Hamiltonian, a metric scale, a coupling strength, or SI units. Here the finite cyclic group and its character space supply all the structure needed for the spectral derivation. This avoids using the larger registry as a substitute for an operator specification.

Verification connection. E03 checks the common-divisibility property of gcd; E04 checks the common-multiple property of lcm. The root-of-unity identification above is a written proof. Exact phase-set checks cover all 14,400 pairs $1 \leq n, m \leq 120$; direct first-return searches cover 1,600 pairs. Fourier Gram matrices were checked for $n = 7, 12, 30$.

5 An operator selects the spectral readout

On $\mathbb{C}^n$ with cyclic indexing, define $(S\psi)_x = \psi_{x-1}$ and

$$L_n = 2I - S - S^*. \quad (9)$$

This shift definition applies for every positive n . At $n = 1$ it gives zero; at $n = 2$ it counts the two shift directions through the same neighbor. It therefore differs from a convention that assigns a simple two-vertex graph a single undirected edge.

Let $u_a(x) = n^{-1/2}e^{2\pi i a x/n}$. Since $Su_a = e^{-2\pi i a/n}u_a$,

$$L_n u_a = \lambda_n(a)u_a, \quad \lambda_n(a) = 2 - 2\cos(2\pi a/n) = 4\sin^2(\pi a/n). \quad (10)$$

The operator is Hermitian and nonnegative. Directly,

$$\langle \psi, L_n \psi \rangle = \sum_{x=0}^{n-1} |\psi_x - \psi_{x-1}|^2 \geq 0. \quad (11)$$

Its zero eigenspace consists of the constant vectors. A uniform reference mode therefore has zero cost under this particular difference operator; changing the operator can change that conclusion.

Every circulant operator has the form $A = \sum_{j=0}^{n-1} a_j S^j$, so the same Fourier basis diagonalizes it, with eigenvalues $\sum_j a_j e^{-2\pi i a j/n}$. Hermiticity imposes the corresponding conjugacy of coefficients and gives real eigenvalues. Cyclic symmetry fixes a useful eigenbasis; it does not fix those eigenvalues. Equation (9) adds a nearest-neighbor difference law to the enumeration.

5.1 From dimensionless differences to a physical ring

Suppose a ring has circumference ℓ , spacing $a = \ell/n$, and a particle model adopts

$$H_n = \frac{\hbar^2}{2Ma^2} L_n + bI. \quad (12)$$

Here M denotes mass and b an energy reference. For fixed mode index j as $n \rightarrow \infty$,

$$\frac{\hbar^2}{2Ma^2} 4\sin^2(\pi j/n) \rightarrow \frac{\hbar^2}{2M} \left(\frac{2\pi j}{\ell} \right)^2. \quad (13)$$

The limit follows from $\sin u/u \rightarrow 1$. It matches the free-particle ring spectrum under periodic boundary conditions [6]. The finite dispersion has a measurable departure from the continuum expression; for $n = 32, j = 1$, the checked ratio equals 0.9967913640.

This supplies a physical realization of the arithmetic-to-operator route. Mass, spacing, the kinetic-energy law, and boundary conditions enter explicitly. A spring ring or an electrical network can use the same difference matrix with a different dynamical equation and calibration. Identical matrices need not mean identical physical energies or frequencies.

Verification connection. The included numerical checks verify $L_n F = F \text{diag}(\lambda)$ for $n = 32$, and diagonalization of a seeded Hermitian circulant. They do not formalize the spectral theorem or the continuum limit in Lean.

6 Energy as residue: three counts and a recoverability theorem

Define the spectral readout for nonzero complex z by

$$\lambda(z) = 2 - z - z^{-1}. \quad (14)$$

For $|z| = 1$, this equals $2 - 2\cos\theta$ with $z = e^{i\theta}$. Inversion preserves the readout: $\lambda(z^{-1}) = \lambda(z)$.

Theorem 6.1 (Exactly two spectral branches). *For nonzero $z, w \in \mathbb{C}$,*

$$\lambda(z) = \lambda(w) \iff z = w \text{ or } zw = 1. \quad (15)$$

Proof. Multiplication by zw gives the identity

$$[\lambda(z) - \lambda(w)]zw = -(z - w)(zw - 1). \quad (16)$$

The complex numbers have no zero divisors, so equality of readouts forces one of the two factors to vanish. Either branch also gives equality by substitution. $\square$

Thus the readout retains an inversion orbit $\{z, z^{-1}\}$, losing orientation unless $z = z^{-1}$. The fixed phases satisfy $z^2 = 1$: $z = 1$ and, when present, $z = -1$. A decoder that recovered every phase from λ would have to assign both z and z^{-1} to the same input for any non-fixed phase. The collision obstruction rules it out. This gives an explicit operator-relative realization of irrecoverable descriptive loss.

Theorem 6.2 (The corrected three-count correspondence). *For positive n, m and $g = \gcd(n, m)$, compare the operators L_n, L_m under the common dimensionless convention (9). Then*

$$N_{\text{phase}} = g, \quad (17)$$

$$N_{\text{distinct}} = \left\lfloor \frac{g}{2} \right\rfloor + 1, \quad (18)$$

$$N_{\text{ordered}} = 2g - \gcd(g, 2). \quad (19)$$

Here the last count enumerates ordered pairs $(a, b) \in C_n \times C_m$ with equal spectral values, including degeneracies.

Proof. The phase count follows from Theorem 4.1. If $z \in \mu_n$ and $w \in \mu_m$ give equal spectral values, Theorem 6.1 gives $z = w$ or $z = w^{-1}$. In either case $z \in \mu_m$, hence $z \in \mu_g$. Therefore the common spectral set equals $\lambda(\mu_g)$.

Inversion on μ_g has $f = \gcd(g, 2)$ fixed points. Its remaining $g - f$ elements occur in pairs, so the number of orbits equals $f + (g - f)/2 = (g + f)/2 = \lfloor g/2 \rfloor + 1$.

For ordered pairs, the positive branch $z = w$ contains g pairs and the negative branch $z = w^{-1}$ contains g pairs. They intersect exactly when $z = w = z^{-1}$, giving f pairs. Inclusion-exclusion yields $2g - f$. $\square$

Example 6.3 (Twelve and eight). For $(n, m) = (12, 8)$, $g = 4$. The common phases are $1, i, -1, -i$; the distinct spectral values are $0, 2, 4$. At value 0, the ordered pair is $(0, 0)$. At value 2, the pairs are $(3, 2), (3, 6), (9, 2), (9, 6)$. At value 4, the pair is $(6, 4)$. Hence the three counts are $4, 3, 6$.

For an odd prime common divisor $g = p$, the counts become $p, (p + 1)/2, 2p - 1$. For coprime moduli they become $1, 1, 1$: only the identity phase and zero Laplacian value survive. When $g = 2$, all common phases are fixed by inversion and the counts coincide at two. These boundary cases expose why a single undifferentiated “shared harmonic count” cannot serve every target.

6.1 What the energy statement adds

The common spectral-value theorem becomes a common-energy theorem when both systems use the same injective calibration

$$E = a\lambda + b, \quad a > 0, \quad (20)$$

with the same a and b . If $a_n \neq a_m$ or $b_n \neq b_m$, physical coincidence requires

$$a_n \lambda_n(j) + b_n = a_m \lambda_m(k) + b_m, \quad (21)$$

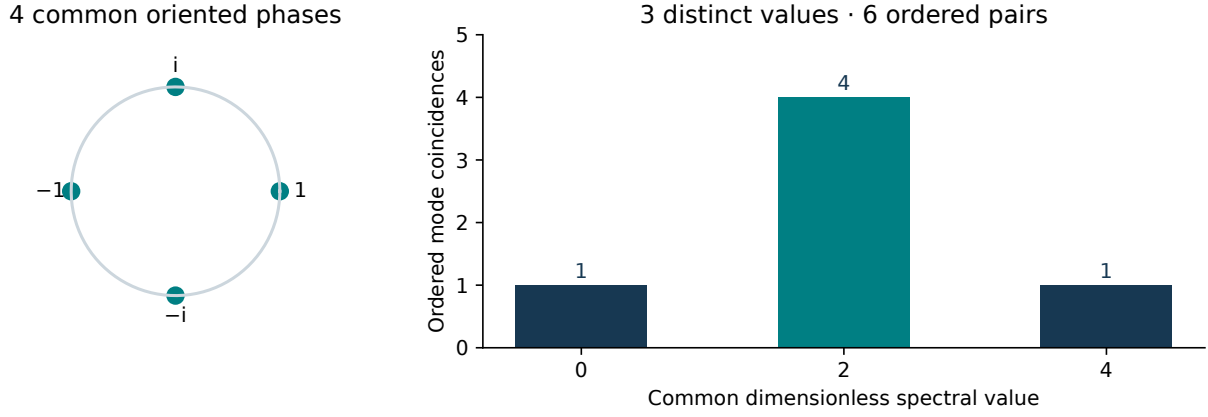


Figure 2: The 12–8 comparison. The spectral operator identifies i with $-i$. Multiplicity records the four cross-frame pairings at their common value 2; it does not restore an orientation decoder from one spectral value.

which the gcd alone does not decide. In the discretized ring example, fixing circumference while changing n changes $a = \hbar^2/(2M(\ell/n)^2)$; one must account for this before comparing energies.

Energy as residue can therefore mean the retained scalar under the operator’s inversion quotient, or the invariant energy expectation under a declared autonomous unitary dynamics. Those are precise claims. Neither identifies energy with an unspecified universal residue independent of operator, state, and calibration.

Verification connection. E07 checks inversion invariance, E08 the forward two-branch implication, and E09 the impossibility of a phase decoder. E10 checks the inclusion-exclusion step conditional on its finite-set cardinality inputs. The root-set and fixed-point counts receive the written proof above and 14,400 exact rational-phase checks; the complete three-count theorem is not encoded end to end in the supplied Lean module.

7 Energy, relative phase, and an operational clock

7.1 Autonomous Hermitian evolution

Adopt a finite-dimensional Hermitian Hamiltonian H and the quantum evolution law

$$U(t) = e^{-iHt/\hbar}, \quad \psi(t) = U(t)\psi(0). \quad (22)$$

Hermiticity gives $U(t)^*U(t) = I$. Since $U(t)$ is a function of H , it commutes with H , and therefore

$$\|\psi(t)\|^2 = \|\psi(0)\|^2, \quad \langle \psi(t), H\psi(t) \rangle = \langle \psi(0), H\psi(0) \rangle. \quad (23)$$

This conserved expectation supplies one operator-relative residue. The proof assumes an autonomous closed model. Driving, environmental exchange, or an explicitly time-dependent Hamiltonian changes the conservation question.

For a common shift $H' = H + bI$,

$$U'(t) = e^{-ibt/\hbar}U(t). \quad (24)$$

The density matrix and every fixed-system observable expectation remain unchanged. Energy gaps satisfy $(E_j + b) - (E_k + b) = E_j - E_k$. The absolute zero shifts while relative phase rates

persist. Coupling the system to another reference requires a consistent common energy convention; independent shifts of coupled blocks can change relative energies.

Rescaling $H \mapsto \alpha H$ and $t \mapsto t/\alpha$, with $\alpha > 0$, preserves Ht and hence U . This covariance shows that dimensionless dynamics alone does not independently fix an energy scale and a time scale. A calibration anchors the numerical interpretation.

7.2 A phase factor alone supplies no internal tick

If $H\psi = E\psi$, then $\psi(t) = e^{-iEt/\hbar}\psi$ and $\rho(t) = \rho(0)$. A time-independent observable consequently has constant expectation. This statement concerns the internal readout of that autonomous eigenstate; it does not remove time from a larger coupled system or from the laboratory description.

For a superposition $\psi(0) = \sum_j c_j |j\rangle$, an observable A gives

$$\langle A \rangle_t = \sum_{j,k} \bar{c}_j c_k A_{jk} e^{i(E_j - E_k)t/\hbar}. \quad (25)$$

A time-varying term requires an energy difference and an observable coherence: $E_j \neq E_k$ and $\bar{c}_j c_k A_{jk} \neq 0$. Merely counting two modes does not guarantee a clock. Degenerate modes can share one energy, and a readout diagonal in energy can miss all relative-phase motion.

For example, take $\psi(0) = (|0\rangle + |1\rangle)/\sqrt{2}$ and $A = |0\rangle\langle 1| + |1\rangle\langle 0|$. Then $\langle A \rangle_t = \cos((E_1 - E_0)t/\hbar)$, with period $h/|E_1 - E_0|$. An observed period therefore fixes a gap once the quantum relation and time calibration are supplied. This is an operational bridge from spectral difference to a clock readout.

Verification connection. E11 checks scalar gap invariance; E12 checks the scalar energy–time rescaling identity. Numerical checks test unitarity, norm and energy conservation, density-matrix shift/scale covariance, and eigenmode stationarity. The matrix-exponential and observable arguments above are written derivations, not additional Lean declarations.

8 Resonance becomes a perturbation prediction

To test whether shared levels matter dynamically, specify the coupling. Let

$$H(\varepsilon) = \begin{pmatrix} H_A & 0 \\ 0 & H_B \end{pmatrix} + \varepsilon \begin{pmatrix} 0 & V \\ V^* & 0 \end{pmatrix} = H_0 + \varepsilon W, \quad (26)$$

with finite Hermitian blocks and real small ε . The purely off-diagonal form of W is essential to the following distinction.

Let P project onto an isolated eigenspace of H_0 at energy E . Substituting $E(\varepsilon) = E + \varepsilon\eta + O(\varepsilon^2)$ and $\psi(\varepsilon) = \psi_0 + O(\varepsilon)$ into the eigenvalue equation and applying P gives

$$PWP\psi_0 = \eta\psi_0. \quad (27)$$

Thus the eigenvalues of PWP supply the first-order shifts. For a level shared between the blocks,

$$PWP = \begin{pmatrix} 0 & V_E \\ V_E^* & 0 \end{pmatrix}, \quad (28)$$

whose nonzero eigenvalues are plus and minus the singular values of V_E . A shared level permits first-order splitting; nonzero coupling within its shared subspace produces it. Selection rules or $V_E = 0$ can remove that splitting.

For a simple level present in only one block, $\langle j|W|j\rangle = 0$. Its first-order shift vanishes, while the second-order coefficient is

$$E_j^{(2)} = \sum_{k \neq j} \frac{|W_{kj}|^2}{E_j - E_k}, \quad (29)$$

assuming a nonzero spectral gap to the other levels. It can vanish by absent coupling or cancellation; “generically second order” does not mean every coefficient must be nonzero.

8.1 A fully specified fresh test

The accompanying program uses

$$H_0 = \text{diag}(0, 2, 0, 3), \quad V = \begin{pmatrix} 1 & 0.3 \\ 0.4 & 0.7 \end{pmatrix}. \quad (30)$$

At the shared zero level, PWP has off-diagonal entries one, giving first-order shifts $\pm\varepsilon$. For the isolated level 2, equation (29) gives $0.4^2/(2-0) + 0.7^2/(2-3) = -0.41$. For the level 3, it gives $0.3^2/(3-0) + 0.7^2/(3-2) = 0.52$.

At $\varepsilon = 0.0003$, direct Hermitian diagonalization returns shared-level slopes -1.0000164990 and 0.9999834990 , and second-order ratios -0.4099999846 and 0.5199999887 . The program also checks $\varepsilon = 0.01, 0.003, 0.001$, recording the convergence toward the predicted coefficients.

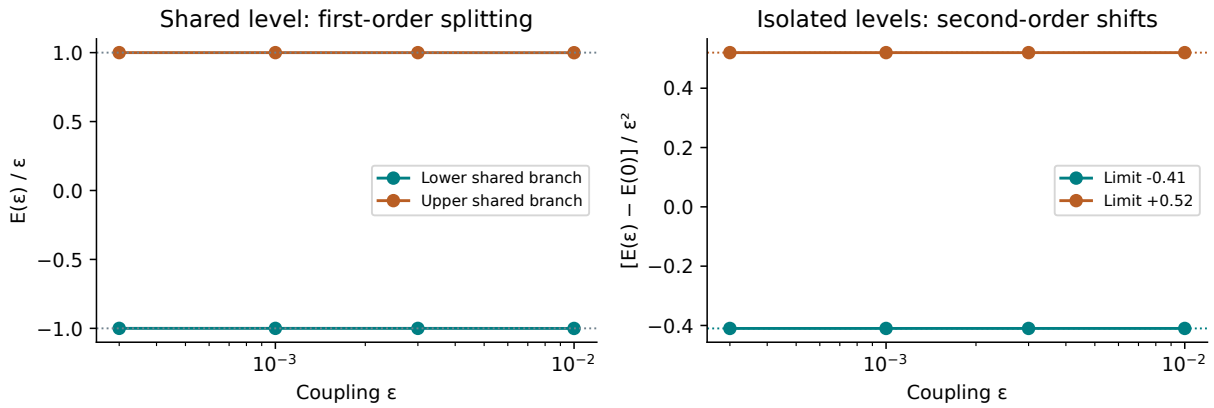


Figure 3: Fresh finite-matrix calculations. Horizontal references show the analytic leading coefficients. All quantities here use the declared model units; no experimental spectrum enters this run.

A physical test can measure uncoupled mode frequencies, establish the energy/frequency mapping, tune a known coupling, and fit the splitting versus its strength. A nonzero linear slope at a purportedly isolated level would challenge the model’s off-diagonal or isolation assumptions. No first-order splitting at a shared level can indicate a vanishing matrix element; the count alone does not force an interaction.

8.2 Address loading and spatial energy density

Assign the canonical integers $0, \dots, N-1$ to M bins by reduction modulo M . Writing $N = qM + r$, $0 \leq r < M$, gives $q+1$ entries in r bins and q in the others. Uniform loading occurs exactly when $M \mid N$.

When $M \nmid N$, this remains a valid map of the selected integer representatives, but it does not descend to a homomorphism $\mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{Z}/M\mathbb{Z}$. Changing a representative by N changes its residue modulo M . Moreover, equal address counts imply equal energy density only after equal energy

per address and the relevant spatial or volume weights have been declared. These conditions keep the energy note's loading result exact at both the combinatorial and physical levels [3].

Verification connection. Fresh checks cover the four perturbation strengths and 1,980 canonical-loading cases. The perturbation expansion and the loading theorem receive written proofs; no Lean declaration here formalizes a full perturbation theory.

9 SI units: the calibration carries physical content

9.1 Seconds, metres, and the propagation relation

The SI fixes the unperturbed caesium-133 ground-state hyperfine transition frequency at

$$\Delta\nu_{\text{Cs}} = 9\,192\,631\,770 \text{ s}^{-1}, \quad (31)$$

and the speed of light in vacuum at

$$c = 299\,792\,458 \text{ m s}^{-1}. \quad (32)$$

These exact defining values establish the second and metre through specified physical references [7, 8]. A realization of those units still has an uncertainty budget: the definition's exact number does not make every clock or optical path perfect.

For a vacuum light path, length and elapsed time satisfy $\ell = c \Delta t$ in the appropriate local inertial description. A round-trip ranging procedure gives $\ell = c \Delta t_{\text{round}}/2$ under its stated stationary-path assumptions. Motion, refractive media, gravity, and instrument delays require the corresponding model corrections. The relation itself demonstrates why a physical length readout cannot omit the timing and propagation convention that gives it meaning.

Theorem 9.1 (Fixed numerical c couples unit rescaling). *Suppose a new length unit equals L old length units and a new time unit equals U old time units, with $L, U > 0$. The new numerical speed is $c' = cU/L$. If $c' = c \neq 0$, then $L = U$.*

Proof. The equality $cU/L = c$ gives $cU = cL$. Cancelling nonzero c yields $U = L$. $\square$

The theorem describes the constraint on unit changes that preserve the same numerical c . It does not prohibit other unit systems; such systems can assign c a different numerical value. It also does not make length and time the same dimension. The conversion factor carries dimensions of speed.

9.2 Joules and dimensionless spectral values

The SI fixes Planck's constant at $h = 6.626\,070\,15 \times 10^{-34} \text{ J s}$, with $\hbar = h/(2\pi)$ [9]. A transition with energy gap ΔE and photon frequency ν satisfies $\Delta E = h\nu$ under the quantum transition model. The dimensionless phase $\Delta E t/\hbar$ then connects energy and time without treating their units as interchangeable labels.

If a model determines only ratios $r_j = E_j/E_*$, specifying an anchor E_* gives $E_j = r_j E_*$. Any two assignments satisfying the same ratios and anchor coincide pointwise. For gap data, use a reference gap ΔE_* ; fixing an additive zero requires additional information. The included Lean proof formalizes uniqueness once the anchor equations are given, not the empirical process of measuring the anchor.

The energy residue therefore needs two specifications: the equivalence under which something remains invariant, and the calibration under which it acquires units. Spectral orientation loss, energy-zero freedom, and dimensional calibration refer to different transformations and must not be merged.

Verification connection. P07 checks the fixed- c rescaling implication; P08 checks uniqueness from calibrated ratios. SI definitions provide the physical reference values. The Lean source does not encode a metrology laboratory or the empirical adequacy of the quantum transition model.

10 Light cones and the impossibility of a one-coordinate description

10.1 The joint interval

Use $T = ct$ so that T and x have the same length units. In $1 + 1$ dimensional Minkowski coordinates, set

$$q(T, x) = x^2 - T^2. \quad (33)$$

For a nonzero separation, $q < 0$ means timelike, $q = 0$ null, and $q > 0$ spacelike. In $3 + 1$ dimensions, replace x^2 by $x^2 + y^2 + z^2$. Future versus past requires a time orientation in addition to the sign of q .

A boost with $|\beta| < 1$, $\beta = v/c$, and $\gamma = (1 - \beta^2)^{-1/2}$ gives

$$T' = \gamma(T - \beta x), \quad x' = \gamma(x - \beta T). \quad (34)$$

Substitution yields

$$x'^2 - T'^2 = \gamma^2[(x - \beta T)^2 - (T - \beta x)^2] \quad (35)$$

$$= \gamma^2(1 - \beta^2)(x^2 - T^2) = x^2 - T^2. \quad (36)$$

The null set and causal classification therefore survive the boost. These are the standard Lorentz relations, stated here to expose exactly how the descriptive dependence works [10].

Theorem 10.1 (Neither projection determines the causal class). *There is no predicate of T alone that agrees with $q(T, x) < 0$ for every (T, x) . There is also no such predicate of x alone.*

Proof. At fixed $T = 1$, $(1, 0)$ is timelike and $(1, 2)$ spacelike. At fixed $x = 0$, $(1, 0)$ is timelike while $(0, 0)$ does not satisfy the strict timelike inequality. Either proposed predicate would assign one truth value to two points requiring different values. One can also avoid the zero separation by using fixed $x = 1$ and comparing $T = 2$ with $T = 0$. $\square$

Theorem 10.2 (A nontrivial boost needs both coordinates). *For $\gamma \neq 0$ and $\beta \neq 0$, no function $f(T)$ equals $\gamma(T - \beta x)$ for all T, x . No function $g(x)$ equals $\gamma(x - \beta T)$ for all T, x .*

Proof. Fix $T = 0$ and compare $x = 0$ with $x = 1$. The boosted time values differ by $-\gamma\beta \neq 0$, while a function of T alone cannot distinguish them. Fixing $x = 0$ and varying T proves the spatial statement. $\square$

The joint interval survives re-description; a discarded coordinate cannot generally be recovered from the retained one. This gives two distinct physical instantiations of the framework: an invariant under an invertible transformation, and an obstruction created by a non-injective projection. A Lorentz boost itself loses no information; applying only one coordinate projection does.

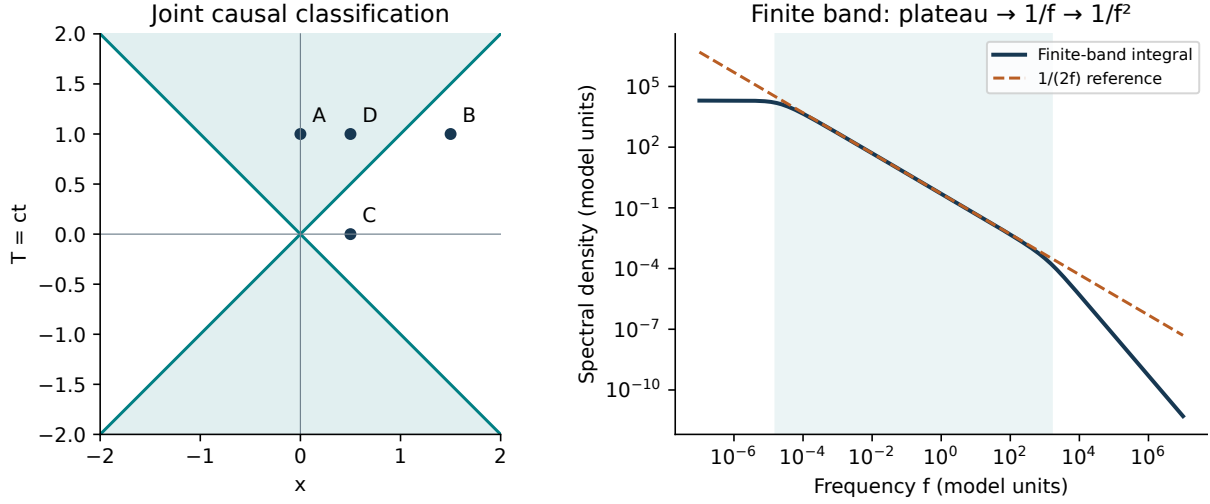


Figure 4: Left: the null lines bound timelike regions. A and B share T but have different causal classes; C and D share x but also differ. Right: the declared finite relaxation band produces a middle $1/f$ region with low- and high-frequency crossovers. The two panels illustrate different physical models, not a derivation of one from the other.

10.2 Finite propagation and local realizations

For causally related events connected by a future-directed signal in the flat model, $\Delta t \geq |\Delta \mathbf{x}|/c$, with equality for vacuum light. Any physical account of a record must specify where and when a preparation, interaction, or readout occurs. Assigning an observer a label does not remove those propagation relations.

The local light cone extends naturally to a Lorentzian metric $g_{\mu\nu}$ through $g_{\mu\nu}v^\mu v^\nu = 0$. The particular linear formulas above apply in the flat model; they do not prove arbitrary global statements about curved spacetime. At an event horizon, local null structure and the global ability to reach an asymptotic region are distinct questions. A coordinate singularity also differs from a physical singularity. For the Schwarzschild example, an appropriate coordinate extension removes the coordinate failure at the horizon without removing the causal boundary [11].

These distinctions matter for the master-key method. A coordinate convention, an information-losing readout, and a global causal restriction require different proofs. The verified nonfactorization results establish the stated coordinate and causal-class obstructions. They do not, by themselves, establish an unrestricted claim about the ontological priority of space or time.

Verification connection. P01–P06 check both boosted-coordinate obstructions, interval preservation, null preservation, and both causal-class obstructions over the real numbers. The algebraic preservation theorem assumes $\gamma^2(1-\beta^2) = 1$; physical boosts additionally select $|\beta| < 1$ and positive γ . Fresh rational-arithmetic checks verify the interval and inverse map in 2,057 cases.

11 Scales, nested traversal, and finite-band predictions

11.1 Planck combinations preserve the same dimensional relation

The conventional Planck combinations satisfy

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}, \quad \ell_P = ct_P. \quad (37)$$

The last equality follows algebraically for positive $c, \hbar, G$. It expresses the same dimensional coupling at that chosen scale. It does not establish that spacetime consists of discrete Planck cells or that all sub-Planck questions lack meaning. Unlike the fixed SI values of c and $\hbar$, G carries experimental uncertainty [9]. Dimensional construction alone does not add a quantum-gravity dynamics.

This separates three operations often compressed into one claim: forming a scale from constants, choosing a resolution for a description, and proving a physical lower bound. Each requires its own premise. The registry's pointwise-versus-uniform distinction provides a useful reminder: positive costs at each tested scale do not automatically produce one positive lower bound across all scales.

11.2 A declared relaxation ladder

Consider independent relaxation contributions with one-sided model spectrum

$$S_\tau(f) = \frac{2A\tau}{1 + (2\pi f\tau)^2}, \quad A > 0, \quad (38)$$

distributed uniformly per logarithmic interval of τ on $[\tau_{\min}, \tau_{\max}]$. Summing uncorrelated contributions means integrating against $d\tau/\tau$. Thus

$$S(f) = 2A \int_{\tau_{\min}}^{\tau_{\max}} \frac{d\tau}{1 + (2\pi f\tau)^2} \quad (39)$$

$$= \frac{A}{\pi f} [\arctan(2\pi f\tau_{\max}) - \arctan(2\pi f\tau_{\min})]. \quad (40)$$

In the middle band $2\pi f\tau_{\min} \ll 1 \ll 2\pi f\tau_{\max}$, the bracket approaches $\pi/2$, giving $S(f) \approx A/(2f)$. At low frequency the spectrum approaches $2A(\tau_{\max} - \tau_{\min})$. At high frequency it approaches

$$S(f) \sim \frac{2A}{(2\pi f)^2} \left(\frac{1}{\tau_{\min}} - \frac{1}{\tau_{\max}} \right). \quad (41)$$

Finite cutoffs therefore predict crossovers, not an exact power law at all frequencies.

The model supplies a concrete nested-scale calculation. Its assumptions include the relaxation kernel, logarithmic weighting, incoherent addition, and the finite range. A different weighting or correlated contributions can change the spectrum. Prime orbit structure does not by itself imply this distribution of physical relaxation times.

The fresh run compares numerical quadrature in $\log \tau$ with (40) at 81 frequencies from 10^{-6} through 10^6 , using $A = 1$, $\tau_{\min} = 10^{-4}$ and $\tau_{\max} = 10^4$ in model units. The maximum relative discrepancy equals 2.4932×10^{-14} against the declared 10^{-9} tolerance. This verifies agreement between two implementations of the model integral. Testing a physical $1/f$ account would additionally require a measured spectrum and independent evidence for the relaxation distribution.

12 Geometric residue without erasing the geometry

The enumerative note also develops closed framed curves and the relation

$$\text{Lk} = \text{Tw} + \text{Wr}. \quad (42)$$

For a smooth closed ribbon with disjoint edges and a consistent framing, the linking number remains invariant under admissible ambient isotopy. Twist and writhe can vary, with changes compensating in their sum. This provides a geometric example of invariant content distributed between description-dependent terms [12].

The qualification on deformation matters. The radial family

$$F_s(x) = x \left[(1 - s) + \frac{sR}{|x|} \right] \quad (43)$$

has radial derivative $1 - s$ away from the origin. For $0 \leq s < 1$, it remains injective on its radial domain and can transport an admissible ribbon through suitable embeddings. At $s = 1$, it collapses every radius onto the sphere and ceases to be injective. One cannot treat that endpoint as an ambient isotopy of an arbitrary knot. In particular, a nontrivial knot cannot become a simple embedded curve lying entirely on a sphere through such an isotopy.

Sphere holonomy supplies a related but separate relation: parallel transport around a suitably oriented loop on a sphere returns a tangent vector rotated by the enclosed solid angle modulo 2π . Its connection to curvature does not identify it with an arbitrary ribbon's twist–writhe exchange. A proposed correspondence must specify the loop, framing, metric, and transport rule.

The energy note's current-loop examples similarly require a declared readout. Symmetry can cancel an ordinary magnetic dipole contribution while preserving a toroidal contribution. Magnetic and toroidal moments carry different dimensions and enter radiation with different frequency factors. Their raw numerical magnitudes cannot be compared as interchangeable measures. A static toroidal pattern alone does not establish an anapole: the radiative cancellation needs the dynamic multipole combination and its frequency-dependent normalization [3].

The source notes report numerical ribbon and current-moment results, but their referenced original programs were not embedded in the supplied PDFs. This edition does not present those historical numbers as a fresh reproduction. Its fresh computations instead have complete programs, specified domains, and captured outputs. The geometric and electromagnetic examples remain valuable because they show exactly where a representation must preserve framing, embedding, and readout information.

13 The master key as a repeatable theorem-and-test method

The resulting method can be stated operationally. First choose the target: an orbit period, oriented phase, energy gap, causal class, or proof conclusion. Then specify which descriptions count as equivalent and which data a readout retains. Prove invariance under the declared equivalence, or exhibit a collision that prevents recovery. Finally add the operator, calibration, and measurement needed for a physical prediction.

Application	Retained relation	Information that must be supplied	Decisive check
Prime traversal	Addition and return under conjugate encoding	Modulus and transported operation	Exact orbit versus divisor characterization
Shared spectra	Inversion class of a phase	Operator convention; multiplicity target	$g, \lfloor g/2 \rfloor + 1, 2g - \gcd(g, 2)$
Energy resonance	Calibrated gaps and coupling	Shared scale/zero; coupling matrix elements	Splitting slope and higher-order shifts
Quantum clock	Relative phase	Coherence, observable, time calibration	Readout oscillation at $\Delta E/\hbar$
Light cone	Joint interval	Both coordinates, metric, time orientation	Interval preservation and collision witnesses
Proof registry	Conclusion under primitive expansion	Formal theory, checker, cost, vocabulary	Conservative expansion and checked certificates

This common structure deserves the name “master key” as a method of disciplined transfer. The same factorization question can expose missing phase orientation or missing spatial data, while a conserved relation can survive a change of coordinate system or ribbon embedding. The strength lies in explicit instantiation: each application receives its own objects, equations, and witnesses.

A general proof registry and a finite prime test also have different termination properties. Testing a fixed integer for divisibility has a finite search bound. Enumerating candidate proofs for an arbitrary formula may continue indefinitely without finding a proof. A successful finite arithmetic experiment does not transfer a universal decision procedure to the registry. Conversely, incompleteness of a sufficiently expressive formal theory does not prevent complete proofs of the finite-group statements developed here.

The current evidence supports a coherent collection of exact mathematical correspondences and physically interpretable models. It supplies reproducible tests of those models and precise conditions for experimental comparison. A universal account of all dynamics would require additional results selecting the operators, interactions, and observed parameter values. Those tasks remain visible because the present method makes their missing inputs explicit.

14 Verification receipts and their exact scope

14.1 Lean acceptance receipt

The combined file `LambdaPartII.lean` was submitted to the public Lean web environment using the `mathlib-stable` project. Its accepted source contains 15 declarations in `EnumerativeEnergy` and eight in `LambdaPhysical`. The physical declarations carry forward the light-cone and calibration work and were checked again in the combined Part II run.

Receipt field	Captured value
UTC capture	2026-09-11T04:23:45.691Z
Lean version reported by evaluation	4.33.0
Project	<code>mathlib-stable</code> (stable release with Mathlib)
Theorem checks / axiom reports	23 / 23
Compiler messages	48: the 46 declaration outputs, version, and completion marker
Errors / warnings	0 / 0 in the accepted run
Admitted-proof dependency	No <code>sorryAx</code> in any printed axiom report
Source identity	Editor clipboard text exactly equal to the submitted source
Completion marker	<code>LAMBDA_PART_II_AUDIT_COMPLETE</code>

This receipt records successful elaboration and kernel checking in the web project; it is not a local `lakebuild` log. The stable project can change over time, and an exact Mathlib commit identifier was not captured. The embedded source-bearing reproduction URL records the submitted source and project; reproducing the historical dependency snapshot would additionally require that project’s exact revision. The full accepted message transcript appears in [Appendix C](#).

The axiom reports show no dependencies for E14; E03–E06 use `propext` and `Quot.sound`; the remaining 18 declarations also use `Classical.choice`. These are the reported logical dependencies. No custom physical axiom or unfinished proof placeholder appears in those reports. Physical interpretation still depends on the explicit model assumptions in the theorem statements and prose.

14.2 Declaration-to-claim map

In the table, E names belong to `EnumerativeEnergy`; P names belong to `LambdaPhysical`. Full types and proofs appear in [Appendix B](#).

ID	Declaration	Formal coverage
E01	<code>prime_iff_no_intermediate_divisor</code>	Primality iff no divisor between 2 and $n - 1$, for $n \geq 2$.
E02	<code>prime_step_gcd_one</code>	Nonzero step below a prime has gcd one.
E03	<code>shared_divisibility</code>	Dividing the gcd iff dividing both inputs.
E04	<code>simultaneous_return</code>	The lcm divides a return index iff both inputs do.
E05	<code>relabeling_preserves_orbit</code>	All iterates commute with a bijective relabeling.
E06	<code>relabeling_preserves_return</code>	The same return indices under transported iteration.
E07	<code>inversion_preserves_spectral_read</code>	$\lambda(z^{-1}) = \lambda(z)$.
E08	<code>spectral_collision_has_two_branches</code>	Equal reads of nonzero complex phases imply equality or reciprocal pairing.
E09	<code>spectral_read_cannot_recover_phase</code>	No universal phase decoder when a non-fixed inversion pair exists.
E10	<code>ordered_count_from_union_data</code>	Union count $2g - f$ from the stated finite-set counts.
E11	<code>common_energy_shift_preserves_gaps</code>	Scalar gap invariance under a common offset.
E12	<code>calibration_rescaling_preserves_phase</code>	$(aE)(t/a) = Et$ for nonzero a .
E13	<code>residue_monotone</code>	Infimum decreases under enlargement with a fixed cost.
E14	<code>conservative_representation_preserves_theorems</code>	Same theorem set under conclusion-preserving expansion and inclusion.
E15	<code>pointwise_positivity_not_uniform</code>	Positive reals admit no uniform positive lower bound.
P01	<code>boosted_time_needs_space</code>	Boosted time cannot factor through time alone for nonzero factors.
P02	<code>boosted_space_needs_time</code>	Boosted space cannot factor through space alone.
P03	<code>lorentz_interval_preserved</code>	Real polynomial interval identity under Lorentz normalization.
P04	<code>lorentz_null_preserved</code>	Null equivalence from the same normalization.
P05	<code>timelike_not_time_only</code>	No predicate of time alone gives all timelike classifications.
P06	<code>timelike_not_space_only</code>	No predicate of space alone gives all timelike classifications.

ID	Declaration	Formal coverage
P07	<code>fixed_c_couples_unit_rescaling</code>	Equal numerical nonzero c forces equal length/time rescaling.
P08	<code>calibrated_ratios_determine_values</code>	Same ratios and anchor determine the same values.

The complete gcd spectral-count theorem, the Fourier spectral theorem, perturbation expansion, quantum evolution, and finite-band integral are not encoded end to end in this Lean module. Their proofs and calculations above state the additional steps. In particular, E10 assumes the branch cardinalities; it does not itself count roots of unity. This distinction prevents a partial formal bridge from being presented as a formalization of the whole physics argument.

14.3 Fresh computational receipt

The Python run completed with status `PASS` at 2026-09-11T04:17:39.474018+00:00, using Python 3.12.14, NumPy 2.3.5 and SciPy 1.17.0. The script independently implements the present checks; it is not a recovered copy of the historical notes' `verify.py`.

Check family	Domain / sample	Observed result
Addition orbit period	19,900 pairs; $1 \leq n < 200, 0 \leq k < n$	Zero failures; explicit traversal versus $n/\gcd(n, k)$.
Prime/full-orbit equivalence	$2 \leq n < 200$; 198 moduli	Zero failures; direct divisor test versus full-orbit property.
Subgroup/divisor correspondence	$1 \leq n < 100$; 99 moduli	Generated subgroup orders and uniqueness agree with divisors.
Base relabeling	7,076 cases; bases 2, 10, 12, 60	Zero failures under transported arithmetic.
Phase/spectral counts	$1 \leq n, m \leq 120$; 14,400 pairs	Exact rational phases; zero failures for sets and all three counts.
Canonical loading	1,980 cases	Uniform counts iff the bin count divides the address count.
First joint return	1,600 pairs, $n, m \leq 40$	Direct search agrees with lcm.
Character orthonormality	$n = 7, 12, 30$	Maximum entry error 3.127×10^{-15} ; tolerance 10^{-12} .
Laplacian eigenvector identity	$n = 32$	Maximum error 6.966×10^{-15} ; tolerance 10^{-10} .
Hermitian circulant	$n = 32$; seed 20260911	Maximum off-diagonal magnitude 2.233×10^{-14} ; tolerance 10^{-10} .
Unitary norm/energy	400 times over $[0, 20]$	Norm error 6.662×10^{-16} ; energy error 1.333×10^{-15} ; both below 10^{-12} .
Energy shift/scale	Two covariance checks	Density errors 1.0592×10^{-16} and numerical zero; tolerance 10^{-11} .
Single-mode stationarity	One Fourier eigenmode	Density error 1.164×10^{-16} ; tolerance 10^{-12} .
Off-diagonal perturbation	Four positive coupling strengths	Smallest- ε slopes within 10^{-4} , second-order coefficients within 10^{-5} .

Check family	Domain / sample	Observed result
Lorentz interval/inverse	2,057 rational cases	Exact equality; zero failures.
Finite-band spectrum	81 logarithmic frequencies	Maximum relative error 2.494×10^{-14} ; tolerance 10^{-9} .

Exact finite enumeration checks the stated domain, not all natural numbers. Floating-point checks compare specified implementations within stated tolerances. Numerical zero means that a particular calculation returned zero, not that floating arithmetic supplies a universal equality proof. The written algebra and formal declarations carry the corresponding universal claims where supplied.

14.4 What a physical test would establish

An experimental continuation can test the calibrated spectral coincidence and splitting predictions, the clock-readout relation, or the finite-band crossover model. Each test needs an apparatus model, independently measured parameters, uncertainty estimates, and the raw data. Agreement would support that physical realization. Disagreement can locate an invalid operator, coupling, calibration, or noise assumption. The invariance method itself helps distinguish those failure points.

Historical source notes refer to tests and demonstrators, including geometric and current-moment calculations. The attached PDFs contain those reports but no embedded scripts or observational data. This edition preserves their conceptual contributions while identifying exactly which results were rerun here. It neither erases the earlier work nor silently upgrades a historical report into a new experimental receipt.

15 Closing result

Primality, cyclic traversal, and character closure fit one exact arithmetic structure. The cyclic Laplacian then produces a controlled quotient of phase information, with the three corrected resonance counts and an explicit impossibility of recovering orientation from energy readout alone. Calibrated dynamics turns those spectral relations into energy gaps, clock signals, and perturbative predictions. Lorentz geometry and SI definitions supply independently explicit physical instances in which joint relations survive while one-coordinate descriptions fail.

The continuation therefore gives the physical and prime work a full theorem-and-test development. Its claims remain attached to their hypotheses, its computations remain reproducible, and its formal receipts identify precisely what Lean checked. Further empirical work can now address particular predictions with the operator, calibration, and readout already stated.

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- [13] Mathlib community. *Mathlib.Data.Nat.Prime.Basic*. [Declaration reference](#).
- [14] Lean web editor. [Live Lean environment](#). Actual Part II run: Lean 4.33.0, `mathlib-stable`, captured 11 September 2026. Exact source-bearing URL embedded separately.

A Reproduction and embedded evidence

The PDF attachment collection contains the combined Lean source, its complete JSON receipt and source-bearing web URL, the Python verification program, the full computational JSON data, the concise standard-output receipt, this paper’s TeX source and figure-generation source, four vector figures, and the three supplied source PDFs. A ZIP attachment, `PartII_reproduction_bundle.zip`, groups the working files with a README for convenient extraction.

To repeat the formal check, open the recorded URL or paste `LambdaPartII.lean` into the named Lean project. Confirm all 23 theorem checks and axiom reports, the reported version, and the final marker. The exact historical Mathlib revision remains an uncaptured dependency detail. To repeat the computational checks, run `python3verify_part2.py` in an environment with NumPy and SciPy. The script writes its new UTC receipt alongside itself; retain the captured edition’s receipt separately when comparing runs.

The following appendices print the entire accepted Lean source, the full displayed Lean message transcript, the computational program, and its concise captured output. Long code lines wrap for display; the embedded UTF-8 files retain the exact original line structure. A document-level SHA value has deliberately not been carried over from an earlier file or presented as self-certification.

B Complete accepted Lean source

```

1 import Mathlib
2
3 /-!
4 Part II: exact arithmetic, spectral projection, and indexed residue.
5 The ledger in the paper gives the scope of each declaration.
6 -/
7 namespace EnumerativeEnergy
8
9 theorem prime_iff_no_intermediate_divisor (n : ℕ) (hn : 2 ≤ n) :
10   Nat.Prime n ↔ ¬ ∃ d : ℕ, d | n ∧ 2 ≤ d ∧ d < n := by
11   constructor
12   · intro hp
13     rintro ⟨d, hd, htwo, hlt⟩
14     have hne : d ≠ 1 := by omega
15     have heq := (hp.dvd_iff_eq hne).mp hd
16     omega
17   · intro h
18     by_contra hnp
19     exact h (Nat.exists_dvd_of_not_prime2 hn hnp)
20
21 theorem prime_step_gcd_one (p k : ℕ) (hp : Nat.Prime p)
22   (hk : 0 < k) (hkp : k < p) : Nat.gcd p k = 1 := by
23   exact (Nat.coprime_of_lt_prime (Nat.ne_of_gt hk) hkp hp).gcd_eq_one
24
25 theorem shared_divisibility (n m d : ℕ) :
26   d | Nat.gcd n m ↔ d | n ∧ d | m := by
27   exact Nat.dvd_gcd_iff
28
29 theorem simultaneous_return (n m t : ℕ) :
30   Nat.lcm n m | t ↔ n | t ∧ m | t := by
31   exact Nat.lcm_dvd_iff
32
33 def orbit {A : Type*} (f : A → A) : ℕ → A → A

```

```

34 | 0, x => x
35 | n + 1, x => f (orbit f n x)
36
37 theorem relabeling_preserves_orbit {A B : Type*}
38   (e : A ≈ B) (f : A → A) (n : ℕ) (x : A) :
39   orbit (fun y => e (f (e.symm y))) n (e x) = e (orbit f n x) := by
40   induction n with
41   | zero => rfl
42   | succ n ih =>
43     simp only [orbit, ih, Equiv.symm_apply_apply]
44
45 theorem relabeling_preserves_return {A B : Type*}
46   (e : A ≈ B) (f : A → A) (n : ℕ) (x : A) :
47   orbit (fun y => e (f (e.symm y))) n (e x) = e x ↔ orbit f n x = x := by
48   rw [relabeling_preserves_orbit]
49   exact e.injective.eq_iff
50
51 noncomputable def spectralRead (z : ℂ) : ℂ := 2 - z - z-1
52
53 theorem inversion_preserves_spectral_read (z : ℂ) :
54   spectralRead z-1 = spectralRead z := by
55   simp only [spectralRead, inv_inv]
56   ring
57
58 theorem spectral_collision_has_two_branches (z w : ℂ)
59   (hz : z ≠ 0) (hw : w ≠ 0) (h : spectralRead z = spectralRead w) :
60   z = w ∨ z * w = 1 := by
61   have hid : (spectralRead z - spectralRead w) * (z * w) =
62     -(z - w) * (z * w - 1) := by
63     dsimp [spectralRead]
64     field_simp
65     ring
66   rw [h, sub_self, zero_mul] at hid
67   have hp : (z - w) * (z * w - 1) = 0 := by
68     linear_combination hid
69   rcases mul_eq_zero.mp hp with heq | hinv
70   · exact 0r.inl (sub_eq_zero.mp heq)
71   · exact 0r.inr (sub_eq_zero.mp hinv)
72
73 theorem spectral_read_cannot_recover_phase (z : ℂ) (hz : z ≠ 0)
74   (hd : z ≠ z-1) :
75   ¬ ∃ f : ℂ → ℂ, ∀ w : ℂ, w ≠ 0 → f (spectralRead w) = w := by
76   rintro {f, hf}
77   have h1 := hf z hz
78   have h2 := hf z-1 (inv_ne_zero hz)
79   rw [inversion_preserves_spectral_read] at h2
80   exact hd (h1.symm.trans h2)
81
82 theorem ordered_count_from_union_data {A : Type*} [DecidableEq A]
83   (s t : Finset A) (g f : ℕ)
84   (hs : s.card = g) (ht : t.card = g) (hi : (s ∩ t).card = f) :
85   (s ∪ t).card = 2 * g - f := by
86   have h := Finset.card_union_add_card_inter s t
87   omega
88
89 theorem common_energy_shift_preserves_gaps (a b offset : ℝ) :
90   (a + offset) - (b + offset) = a - b := by ring
91
92 theorem calibration_rescaling_preserves_phase (scale energy time : ℝ)

```

```

93     (hs : scale ≠ 0) :
94     (scale * energy) * (time / scale) = energy * time := by
95     field_simp
96
97 noncomputable def indexedResidue {A : Type*} (S : Set A) (cost : A → ENNReal) : ENNReal :=
98     sInf (cost '' S)
99
100 theorem residue_monotone {A : Type*} (S T : Set A)
101     (cost : A → ENNReal) (h : S ⊆ T) :
102     indexedResidue T cost ≤ indexedResidue S cost := by
103     unfold indexedResidue
104     apply sInf_le_sInf
105     rintro y (x, hx, rfl)
106     exact (x, h hx, rfl)
107
108 theorem conservative_representation_preserves_theorems {P T : Type*}
109     (R0 R1 : Set P) (conclusion : P → T) (expand : P → P)
110     (hin : R0 ⊆ R1)
111     (hex : ∀ p ∈ R1, expand p ∈ R0 ∧ conclusion (expand p) = conclusion p)
112     (target : T) :
113     (∃ p ∈ R0, conclusion p = target) ↔ (∃ p ∈ R1, conclusion p = target) := by
114     constructor
115     · rintro (p, hp, hc)
116     exact (p, hin hp, hc)
117     · rintro (p, hp, hc)
118     obtain (he, ht) := hex p hp
119     exact (expand p, he, ht.trans hc)
120
121 theorem pointwise_positivity_not_uniform :
122     (∀ x : ℝ, 0 < x → 0 < x) ∧
123     ¬ ∃ epsilon : ℝ, 0 < epsilon ∧ ∀ x : ℝ, 0 < x → epsilon ≤ x := by
124     constructor
125     · intro x hx
126     exact hx
127     · rintro (epsilon, he, hbound)
128     have hhalf : 0 < epsilon / 2 := by linarith
129     have hh := hbound (epsilon / 2) hhalf
130     linarith
131
132 end EnumerativeEnergy
133
134 namespace LambdaPhysical
135
136 def boostTime (g b T x : ℝ) : ℝ := g * (T - b * x)
137 def boostSpace (g b T x : ℝ) : ℝ := g * (x - b * T)
138 def interval (T x : ℝ) : ℝ := x ^ 2 - T ^ 2
139
140 theorem boosted_time_needs_space (g b : ℝ) (hg : g ≠ 0) (hb : b ≠ 0) :
141     ¬ ∃ f : ℝ → ℝ, ∀ T x, f T = boostTime g b T x := by
142     rintro (f, hf)
143     have h0 := hf 0 0
144     have h1 := hf 0 1
145     simp only [boostTime, mul_zero, sub_zero] at h0
146     simp only [boostTime, mul_one, zero_sub] at h1
147     have hgb : g * b = 0 := by nlinarith
148     exact (mul_ne_zero hg hb) hgb
149
150 theorem boosted_space_needs_time (g b : ℝ) (hg : g ≠ 0) (hb : b ≠ 0) :
151     ¬ ∃ f : ℝ → ℝ, ∀ T x, f x = boostSpace g b T x := by

```

```

152   rintro {f, hf}
153   have h0 := hf 0 0
154   have h1 := hf 1 0
155   simp only [boostSpace, mul_zero, sub_zero] at h0
156   simp only [boostSpace, mul_one, zero_sub] at h1
157   have hgb : g * b = 0 := by nlinarith
158   exact (mul_ne_zero hg hb) hgb
159
160   theorem lorentz_interval_preserved (g b T x : ℝ)
161     (hn : g ^ 2 * (1 - b ^ 2) = 1) :
162     interval (boostTime g b T x) (boostSpace g b T x) = interval T x := by
163     unfold interval boostTime boostSpace
164     calc
165       (g * (x - b * T)) ^ 2 - (g * (T - b * x)) ^ 2 =
166         (g ^ 2 * (1 - b ^ 2)) * (x ^ 2 - T ^ 2) := by ring
167       _ = x ^ 2 - T ^ 2 := by rw [hn]; ring
168
169   theorem lorentz_null_preserved (g b T x : ℝ)
170     (hn : g ^ 2 * (1 - b ^ 2) = 1) :
171     (interval (boostTime g b T x) (boostSpace g b T x) = 0) ↔
172     interval T x = 0 := by
173     rw [lorentz_interval_preserved g b T x hn]
174
175   theorem timelike_not_time_only :
176     ¬ ∃ P : ℝ → Prop, ∀ T x, P T ↔ interval T x < 0 := by
177     rintro {P, h}
178     have hp : P 1 := (h 1 0).mpr (by norm_num [interval])
179     have hn := (h 1 2).mp hp
180     norm_num [interval] at hn
181
182   theorem timelike_not_space_only :
183     ¬ ∃ P : ℝ → Prop, ∀ T x, P x ↔ interval T x < 0 := by
184     rintro {P, h}
185     have hp : P 1 := (h 2 1).mpr (by norm_num [interval])
186     have hn := (h 0 1).mp hp
187     norm_num [interval] at hn
188
189   theorem fixed_c_couples_unit_rescaling (c L U : ℝ)
190     (hc : c ≠ 0) (hL : L ≠ 0) (h : c * U / L = c) : L = U := by
191     have he : c * U = c * L := (div_eq_iff hL).mp h
192     exact (mul_left_cancel₀ hc he).symm
193
194   theorem calibrated_ratios_determine_values {I : Type*}
195     (r E F : I → ℝ) (anchor : ℝ)
196     (hE : ∀ i, E i = r i * anchor)
197     (hF : ∀ i, F i = r i * anchor) : E = F := by
198     funext i
199     exact (hE i).trans (hF i).symm
200
201
202   end LambdaPhysical
203
204   #check EnumerativeEnergy.prime_iff_no_intermediate_divisor
205   #check EnumerativeEnergy.prime_step_gcd_one
206   #check EnumerativeEnergy.shared_divisibility
207   #check EnumerativeEnergy.simultaneous_return
208   #check EnumerativeEnergy.relabeling_preserves_orbit
209   #check EnumerativeEnergy.relabeling_preserves_return
210   #check EnumerativeEnergy.inversion_preserves_spectral_read

```

```

211 #check EnumerativeEnergy.spectral_collision_has_two_branches
212 #check EnumerativeEnergy.spectral_read_cannot_recover_phase
213 #check EnumerativeEnergy.ordered_count_from_union_data
214 #check EnumerativeEnergy.common_energy_shift_preserves_gaps
215 #check EnumerativeEnergy.calibration_rescaling_preserves_phase
216 #check EnumerativeEnergy.residue_monotone
217 #check EnumerativeEnergy.conservative_representation_preserves_theorems
218 #check EnumerativeEnergy.pointwise_positivity_not_uniform
219 #check LambdaPhysical.boosted_time_needs_space
220 #check LambdaPhysical.boosted_space_needs_time
221 #check LambdaPhysical.lorentz_interval_preserved
222 #check LambdaPhysical.lorentz_null_preserved
223 #check LambdaPhysical.timelike_not_time_only
224 #check LambdaPhysical.timelike_not_space_only
225 #check LambdaPhysical.fixed_c_couples_unit_rescaling
226 #check LambdaPhysical.calibrated_ratios_determine_values
227 #print axioms EnumerativeEnergy.prime_iff_no_intermediate_divisor
228 #print axioms EnumerativeEnergy.prime_step_gcd_one
229 #print axioms EnumerativeEnergy.shared_divisibility
230 #print axioms EnumerativeEnergy.simultaneous_return
231 #print axioms EnumerativeEnergy.relabeling_preserves_orbit
232 #print axioms EnumerativeEnergy.relabeling_preserves_return
233 #print axioms EnumerativeEnergy.inversion_preserves_spectral_read
234 #print axioms EnumerativeEnergy.spectral_collision_has_two_branches
235 #print axioms EnumerativeEnergy.spectral_read_cannot_recover_phase
236 #print axioms EnumerativeEnergy.ordered_count_from_union_data
237 #print axioms EnumerativeEnergy.common_energy_shift_preserves_gaps
238 #print axioms EnumerativeEnergy.calibration_rescaling_preserves_phase
239 #print axioms EnumerativeEnergy.residue_monotone
240 #print axioms EnumerativeEnergy.conservative_representation_preserves_theorems
241 #print axioms EnumerativeEnergy.pointwise_positivity_not_uniform
242 #print axioms LambdaPhysical.boosted_time_needs_space
243 #print axioms LambdaPhysical.boosted_space_needs_time
244 #print axioms LambdaPhysical.lorentz_interval_preserved
245 #print axioms LambdaPhysical.lorentz_null_preserved
246 #print axioms LambdaPhysical.timelike_not_time_only
247 #print axioms LambdaPhysical.timelike_not_space_only
248 #print axioms LambdaPhysical.fixed_c_couples_unit_rescaling
249 #print axioms LambdaPhysical.calibrated_ratios_determine_values
250 #eval Lean.versionString
251 #eval "LAMBDA_PART_II_AUDIT_COMPLETE"

```

C Complete Lean message transcript

The initial “No info found” text refers to the editor cursor’s empty position after the file; it is not a compiler error. The “All Messages (48)” collection below contains the acceptance evidence.

```

mathlib-stable.lean:252:0
No info found.
All Messages (48)
mathlib-stable.lean:204:0
EnumerativeEnergy.prime_iff_no_intermediate_divisor (n : ℕ) (hn : 2 ≤ n) : Nat.Prime n ↔ ¬∃ d, d | n
  ↔ 1 < d & d < n
mathlib-stable.lean:205:0
EnumerativeEnergy.prime_step_gcd_one (p k : ℕ) (hp : Nat.Prime p) (hk : 0 < k) (hkp : k < p) : p.gcd
  ↔ k = 1
mathlib-stable.lean:206:0
EnumerativeEnergy.shared_divisibility (n m d : ℕ) : d | n.gcd m ↔ d | n & d | m
mathlib-stable.lean:207:0
EnumerativeEnergy.simultaneous_return (n m t : ℕ) : n.lcm m | t ↔ n | t & m | t
mathlib-stable.lean:208:0
EnumerativeEnergy.relabeling_preserves_orbit.{u_1, u_2} {A : Type u_1} {B : Type u_2} (e : A ≈ B) (f
  ↔ : A → A) (n : ℕ)
  (x : A) : EnumerativeEnergy.orbit (fun y => e (f (e.symm y))) n (e x) = e (EnumerativeEnergy.orbit
  ↔ f n x)
mathlib-stable.lean:209:0
EnumerativeEnergy.relabeling_preserves_return.{u_1, u_2} {A : Type u_1} {B : Type u_2} (e : A ≈ B)
  ↔ (f : A → A) (n : ℕ)
  (x : A) : EnumerativeEnergy.orbit (fun y => e (f (e.symm y))) n (e x) = e x ↔
  ↔ EnumerativeEnergy.orbit f n x = x
mathlib-stable.lean:210:0
EnumerativeEnergy.inversion_preserves_spectral_read (z : ℂ) :
  EnumerativeEnergy.spectralRead z-1 = EnumerativeEnergy.spectralRead z
mathlib-stable.lean:211:0
EnumerativeEnergy.spectral_collision_has_two_branches (z w : ℂ) (hz : z ≠ 0) (hw : w ≠ 0)
  (h : EnumerativeEnergy.spectralRead z = EnumerativeEnergy.spectralRead w) : z = w ∨ z * w = 1
mathlib-stable.lean:212:0
EnumerativeEnergy.spectral_read_cannot_recover_phase (z : ℂ) (hz : z ≠ 0) (hd : z ≠ z-1) :
  ¬∃ f, ∀ (w : ℂ), w ≠ 0 → f (EnumerativeEnergy.spectralRead w) = w
mathlib-stable.lean:213:0
EnumerativeEnergy.ordered_count_from_union_data.{u_1} {A : Type u_1} [DecidableEq A] (s t : Finset
  ↔ A) (g f : ℕ)
  (hs : s.card = g) (ht : t.card = g) (hi : (s ∩ t).card = f) : (s ∪ t).card = 2 * g - f
mathlib-stable.lean:214:0
EnumerativeEnergy.common_energy_shift_preserves_gaps (a b offset : ℝ) : a + offset - (b + offset) =
  ↔ a - b
mathlib-stable.lean:215:0
EnumerativeEnergy.calibration_rescaling_preserves_phase (scale energy time : ℝ) (hs : scale ≠ 0) :
  scale * energy * (time / scale) = energy * time
mathlib-stable.lean:216:0
EnumerativeEnergy.residue_monotone.{u_1} {A : Type u_1} (S T : Set A) (cost : A → ENNReal) (h : S ⊆
  ↔ T) :
  EnumerativeEnergy.indexedResidue T cost ≤ EnumerativeEnergy.indexedResidue S cost
mathlib-stable.lean:217:0
EnumerativeEnergy.conservative_representation_preserves_theorems.{u_1, u_2} {P : Type u_1} {T :
  ↔ Type u_2}
  (R0 R1 : Set P) (conclusion : P → T) (expand : P → P) (hin : R0 ⊆ R1)
  (hex : ∀ p ∈ R1, expand p ∈ R0 & conclusion (expand p) = conclusion p) (target : T) :
  (∃ p ∈ R0, conclusion p = target) ↔ ∃ p ∈ R1, conclusion p = target
mathlib-stable.lean:218:0

```

```

EnumerativeEnergy.pointwise_positivity_not_uniform :
  (∀ (x : ℝ), 0 < x → 0 < x) ∧ ¬∃ epsilon, 0 < epsilon ∧ ∀ (x : ℝ), 0 < x → epsilon ≤ x
mathlib-stable.lean:219:0
LambdaPhysical.boosted_time_needs_space (g b : ℝ) (hg : g ≠ 0) (hb : b ≠ 0) :
  ¬∃ f, ∀ (T x : ℝ), f T = LambdaPhysical.boostTime g b T x
mathlib-stable.lean:220:0
LambdaPhysical.boosted_space_needs_time (g b : ℝ) (hg : g ≠ 0) (hb : b ≠ 0) :
  ¬∃ f, ∀ (T x : ℝ), f x = LambdaPhysical.boostSpace g b T x
mathlib-stable.lean:221:0
LambdaPhysical.lorentz_interval_preserved (g b T x : ℝ) (hn : g ^ 2 * (1 - b ^ 2) = 1) :
  LambdaPhysical.interval (LambdaPhysical.boostTime g b T x) (LambdaPhysical.boostSpace g b T x) =
    LambdaPhysical.interval T x
mathlib-stable.lean:222:0
LambdaPhysical.lorentz_null_preserved (g b T x : ℝ) (hn : g ^ 2 * (1 - b ^ 2) = 1) :
  LambdaPhysical.interval (LambdaPhysical.boostTime g b T x) (LambdaPhysical.boostSpace g b T x) = 0
  ⇔
    LambdaPhysical.interval T x = 0
mathlib-stable.lean:223:0
LambdaPhysical.timelike_not_time_only : ¬∃ P, ∀ (T x : ℝ), P T ⇔ LambdaPhysical.interval T x < 0
mathlib-stable.lean:224:0
LambdaPhysical.timelike_not_space_only : ¬∃ P, ∀ (T x : ℝ), P x ⇔ LambdaPhysical.interval T x < 0
mathlib-stable.lean:225:0
LambdaPhysical.fixed_c_couples_unit_rescaling (c L U : ℝ) (hc : c ≠ 0) (hL : L ≠ 0) (h : c * U / L =
  ⇔ c) : L = U
mathlib-stable.lean:226:0
LambdaPhysical.calibrated_ratios_determine_values.{u_1} {I : Type u_1} (r E F : I → ℝ) (anchor : ℝ)
  (hE : ∀ (i : I), E i = r i * anchor) (hF : ∀ (i : I), F i = r i * anchor) : E = F
mathlib-stable.lean:227:0
'EnumerativeEnergy.prime_iff_no_intermediate_divisor' depends on axioms: [propext, Classical.choice,
  ⇔ Quot.sound]
mathlib-stable.lean:228:0
'EnumerativeEnergy.prime_step_gcd_one' depends on axioms: [propext, Classical.choice, Quot.sound]
mathlib-stable.lean:229:0
'EnumerativeEnergy.shared_divisibility' depends on axioms: [propext, Quot.sound]
mathlib-stable.lean:230:0
'EnumerativeEnergy.simultaneous_return' depends on axioms: [propext, Quot.sound]
mathlib-stable.lean:231:0
'EnumerativeEnergy.relabeling_preserves_orbit' depends on axioms: [propext, Quot.sound]
mathlib-stable.lean:232:0
'EnumerativeEnergy.relabeling_preserves_return' depends on axioms: [propext, Quot.sound]
mathlib-stable.lean:233:0
'EnumerativeEnergy.inversion_preserves_spectral_read' depends on axioms: [propext, Classical.choice,
  ⇔ Quot.sound]
mathlib-stable.lean:234:0
'EnumerativeEnergy.spectral_collision_has_two_branches' depends on axioms: [propext,
  ⇔ Classical.choice, Quot.sound]
mathlib-stable.lean:235:0
'EnumerativeEnergy.spectral_read_cannot_recover_phase' depends on axioms: [propext,
  ⇔ Classical.choice, Quot.sound]
mathlib-stable.lean:236:0
'EnumerativeEnergy.ordered_count_from_union_data' depends on axioms: [propext, Classical.choice,
  ⇔ Quot.sound]
mathlib-stable.lean:237:0
'EnumerativeEnergy.common_energy_shift_preserves_gaps' depends on axioms: [propext,
  ⇔ Classical.choice, Quot.sound]
mathlib-stable.lean:238:0
'EnumerativeEnergy.calibration_rescaling_preserves_phase' depends on axioms: [propext,
  ⇔ Classical.choice, Quot.sound]
mathlib-stable.lean:239:0

```

```

'EnumerativeEnergy.residue_monotone' depends on axioms: [propext, Classical.choice, Quot.sound]
mathlib-stable.lean:240:0
'EnumerativeEnergy.conservative_representation_preserves_theorems' does not depend on any axioms
mathlib-stable.lean:241:0
'EnumerativeEnergy.pointwise_positivity_not_uniform' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
mathlib-stable.lean:242:0
'LambdaPhysical.boosted_time_needs_space' depends on axioms: [propext, Classical.choice, Quot.sound]
mathlib-stable.lean:243:0
'LambdaPhysical.boosted_space_needs_time' depends on axioms: [propext, Classical.choice, Quot.sound]
mathlib-stable.lean:244:0
'LambdaPhysical.lorentz_interval_preserved' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
mathlib-stable.lean:245:0
'LambdaPhysical.lorentz_null_preserved' depends on axioms: [propext, Classical.choice, Quot.sound]
mathlib-stable.lean:246:0
'LambdaPhysical.timelike_not_time_only' depends on axioms: [propext, Classical.choice, Quot.sound]
mathlib-stable.lean:247:0
'LambdaPhysical.timelike_not_space_only' depends on axioms: [propext, Classical.choice, Quot.sound]
mathlib-stable.lean:248:0
'LambdaPhysical.fixed_c_couples_unit_rescaling' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
mathlib-stable.lean:249:0
'LambdaPhysical.calibrated_ratios_determine_values' depends on axioms: [propext, Classical.choice,
↪ Quot.sound]
mathlib-stable.lean:250:0
"4.33.0"
mathlib-stable.lean:251:0
"LAMBDA_PART_II_AUDIT_COMPLETE"
Restart File

```

D Computational verification program

```

1  """Exact finite-domain checks and numerical model checks for Lambda Part II.
2  Run: python3 verify_part2.py
3  Dependencies: numpy, scipy. No observational dataset is loaded.
4  """
5  from pathlib import Path
6  from fractions import Fraction as Q
7  from collections import Counter
8  from datetime import datetime, timezone
9  from math import gcd, lcm, pi, atan2, log, exp
10 import json, platform
11 import numpy as np
12 import scipy
13 from scipy.linalg import expm
14 from scipy.integrate import quad
15
16 OUT = Path(__file__).resolve().parent
17 records = []
18 def record(name, cases, **data):
19     records.append(dict(name=name, cases=cases, status='PASS', **data))
20
21 # Explicit addition orbits; the expected gcd is not used to generate them.
22 orbit_rows=[]
23 for n in range(1,200):
24     for k in range(n):
25         x=0; seen=[]
26         while True:
27             seen.append(x);x=(x+k)%n
28             if x==0:break
29             assert x not in seen
30             assert len(seen)==n//gcd(n,k)
31             orbit_rows.append([n,k,len(seen)])
32 record('addition_orbit_period',len(orbit_rows),domain='1<=n<200; 0<=k<n',failures=0)
33
34 def prime(n):
35     return n>=2 and all(n%d for d in range(2,n))
36 for n in range(2,200):
37     allfull=all(row[2]==n for row in orbit_rows if row[0]==n and row[1]>0)
38     assert allfull==prime(n)
39 record('prime_full_orbit_equivalence',198,domain='2<=n<200',failures=0)
40 for n in range(1,100):
41     subgroups={frozenset((j*k)%n for j in range(n)) for k in range(n)}
42     orders={len(s) for s in subgroups}
43     divisors={d for d in range(1,n+1) if n%d==0}
44     assert orders==divisors and len(subgroups)==len(divisors)
45 record('subgroup_divisor_correspondence',99,domain='1<=n<100',failures=0)
46
47 # Relabel a numeric state in bases 2,10,12,60, and transport its operation.
48 def digits(n,b):
49     out=[]
50     while n:out.append(n%b);n//=b
51     return tuple(reversed(out or [0]))
52 def value(ds,b):
53     v=0
54     for d in ds:v=v*b+d
55     return v
56 base_cases=0

```

```

57 for b in [2,10,12,60]:
58     for n in range(2,60):
59         for k in range(n):
60             x=0;steps=0
61             while True:
62                 encoded=digits(x,b)
63                 x=(value(encoded,b)+k)%n;steps+=1
64                 if x==0:break
65             assert steps==n//gcd(n,k);base_cases+=1
66 record('base_relabeling_preserves_period',base_cases,bases=[2,10,12,60],failures=0)
67
68 # Exact phases in Q/Z and exact inversion orbits. No floating cosine comparison.
69 phases={n:{Q(k,n) for k in range(n)} for n in range(1,121)}
70 fold=lambda q:min(q,(-q)%1)
71 spectra={n:Counter(fold(q) for q in phases[n]) for n in phases}
72 pair_rows=[]
73 for n in phases:
74     for m in phases:
75         g=gcd(n,m)
76         common=phases[n]&phases[m]
77         shared=spectra[n].keys()&spectra[m].keys()
78         ordered=sum(spectra[n][q]*spectra[m][q] for q in shared)
79         assert common==phases[g]
80         assert shared==spectra[g].keys()
81         assert len(common)==g and len(shared)==g//2+1
82         assert ordered==2*g-gcd(g,2)
83         pair_rows.append([n,m,g,len(common),len(shared),ordered])
84 record('phase_and_spectral_counts',len(pair_rows),domain='1<=n,m<=120',
85       arithmetic='fractions.Fraction',failures=0,
86       example_12_8=[r for r in pair_rows if r[:2]==[12,8]][0])
87
88 loading=0;returns=0
89 for N in range(1,100):
90     for M in range(1,21):
91         counts=[sum(a%M==j for a in range(N)) for j in range(M)]
92         assert (len(set(counts))==1)==(N%M==0);loading+=1
93 for n in range(1,41):
94     for m in range(1,41):
95         ret=next(t for t in range(1,n*m+1) if t%n==0 and t%m==0)
96         assert ret==lcm(n,m);returns+=1
97 record('canonical_address_loading',loading,failures=0)
98 record('first_simultaneous_return',returns,failures=0)
99
100 # DFT, cyclic Laplacian, and autonomous finite Hermitian dynamics.
101 gram=[]
102 for n in [7,12,30]:
103     F=np.exp(2j*pi*np.outer(np.arange(n),np.arange(n))/n)/np.sqrt(n)
104     err=float(np.max(np.abs(F.conj().T@F-np.eye(n))))
105     assert err<1e-12;gram.append([n,err])
106 record('character_orthonormality',3,tolerance=1e-12,errors=gram)
107 n=32;F=np.exp(2j*pi*np.outer(np.arange(n),np.arange(n))/n)/np.sqrt(n)
108 S=np.roll(np.eye(n),1,axis=0);L=2*np.eye(n)-S-S.T
109 lam=2-2*np.cos(2*pi*np.arange(n)/n)
110 eigerr=float(np.max(np.abs(L@F-F*lam)))
111 assert eigerr<1e-10
112 record('laplacian_character_identity',n,tolerance=1e-10,max_error=eigerr,
113       low_mode_dispersion_ratio=float(lam[1]/(2*pi/n)**2))
114 rng=np.random.default_rng(20260911)
115 raw=rng.normal(size=n)+1j*rng.normal(size=n)

```

```

116 C=sum(raw[k]*np.linalg.matrix_power(S,k) for k in range(n))
117 H=(C+C.conj()).T)/2
118 diagonal=F.conj().T@H@F
119 off=float(np.max(np.abs(diagonal-np.diag(np.diag(diagonal)))))
120 assert off<1e-10
121 record('hermitian_circulant_diagonalization',1,max_off_diagonal=off,tolerance=1e-10,seed=20260911)
122
123 psi=rng.normal(size=n)+1j*rng.normal(size=n);psi/=np.linalg.norm(psi)
124 energies,U=np.linalg.eigh(L);coeff=U.conj().T@psi
125 E0=float(np.vdot(psi,L@psi).real)
126 maxnorm=maxenergy=0.0
127 for t in np.linspace(0,20,400):
128     state=U@(np.exp(-1j*energies*t)*coeff)
129     maxnorm=max(maxnorm,abs(float(np.vdot(state,state).real)-1))
130     maxenergy=max(maxenergy,abs(float(np.vdot(state,L@state).real)-E0))
131 assert maxnorm<1e-12 and maxenergy<1e-12
132 t=.73;alpha=3.7;offset=2.3
133 Ut=expm(-1j*L*t)
134 unitarity=float(np.max(np.abs(Ut.conj().T@Ut-np.eye(n))))
135 state=Ut@psi
136 shifted=expm(-1j*(L+offset*np.eye(n))*t)@psi
137 scaled=expm(-1j*(alpha*L)*(t/alpha))@psi
138 density=lambda v:np.outer(v,v.conj())
139 shift_error=float(np.max(np.abs(density(state)-density(shifted))))
140 scale_error=float(np.max(np.abs(density(state)-density(scaled))))
141 assert max(unitarity,shift_error,scale_error)<1e-11
142 record('unitary_norm_and_energy',400,max_norm_error=maxnorm,max_energy_error=maxenergy,
143       max_unitarity_error=unitarity,norm_energy_tolerance=1e-12,unitarity_tolerance=1e-11)
144 record('energy_shift_and_scale_covariance',2,density_shift_error=shift_error,
145       density_scale_error=scale_error,tolerance=1e-11)
146 eigen=F[:,3];stationary=expm(-1j*L*t)@eigen
147 stat_error=float(np.max(np.abs(density(eigen)-density(stationary))))
148 assert stat_error<1e-12
149 record('single_mode_density_stationarity',1,density_error=stat_error,tolerance=1e-12)
150
151 # First/second-order splitting under explicitly off-diagonal coupling.
152 H0=np.diag([0.,2.,0.,3.]);V=np.array([[1.,.3],[.4,.7]])
153 W=np.block([[np.zeros((2,2)),V],[V.T,np.zeros((2,2))]])
154 pert=[]
155 for eps in [1e-2,3e-3,1e-3,3e-4]:
156     vals=np.linalg.eigvalsh(H0+eps*W)
157     pert.append(dict(epsilon=eps,shared_slopes=(vals[2]/eps).tolist(),
158                    level2_second_order=float((vals[2]-2)/eps**2),
159                    level3_second_order=float((vals[3]-3)/eps**2)))
160 last=pert[-1]
161 assert max(abs(last['shared_slopes'][0]+1),abs(last['shared_slopes'][1]-1))<1e-4
162 assert abs(last['level2_second_order']+.41)<1e-5
163 assert abs(last['level3_second_order']-.52)<1e-5
164 record('off_diagonal_perturbation',4,values=pert,shared_slope_tolerance=1e-4,
165       second_order_coefficient_tolerance=1e-5)
166
167 # Exact Lorentz bridge and finite-band relaxation spectrum.
168 lorentz=0
169 for k in range(-8,9):
170     u=Q(k,10);beta=2*u/(1+u*u);gamma=(1+u*u)/(1-u*u)
171     for T in range(-5,6):
172         for x in range(-5,6):
173             Tp,xp=gamma*(T-beta*x),gamma*(x-beta*T)
174             assert xp*xp-Tp*Tp==x*x-T*T

```

```

175         assert (gamma*(Tp+beta*xp),gamma*(xp+beta*Tp))== (T,x)
176         lorentz+=1
177     record('lorentz_interval_and_inverse',lorentz,arithmetic='fractions.Fraction',failures=0)
178     spectral=[];lo=1e-4;hi=1e4
179     for f in np.logspace(-6,6,81):
180         omega=2*pi*f
181         calc,estimate=quad(lambda s:2*exp(s)/(1+(omega*exp(s))**2),log(lo),log(hi),
182                             epsabs=1e-20,epsrel=1e-11,limit=300)
183         closed=atan2(omega*(hi-lo),1+omega**2*hi*lo)/(pi*f)
184         rel=abs(calc-closed)/closed
185         assert rel<1e-9
186         spectral.append(dict(f=float(f),quadrature=calc,closed=closed,
187                             relative_error=rel,error_estimate=estimate))
188     record('finite_band_spectrum',81,tolerance=1e-9,
189           max_relative_error=max(r['relative_error'] for r in spectral))
190
191     result=dict(status='PASS',capture_time_utc=datetime.now(timezone.utc).isoformat(),
192                python=platform.python_version(),numpy=np.__version__,scipy=scipy.__version__,
193                scope='Exhaustive stated finite domains and numerical mathematical-model checks; no
194                ↪ observational dataset.',
195                checks=records,orbit_rows=orbit_rows,phase_spectrum_rows=pair_rows,
196                spectrum=spectral,perturbation=pert)
197     (OUT/'PART_II_COMPUTATIONAL_RECEIPT.json').write_text(json.dumps(result,indent=2)+'\n')
198     print(json.dumps({k:v for k,v in result.items() if k not in
199                       ↪ ['orbit_rows','phase_spectrum_rows','spectrum','perturbation']},indent=2))

```

E Captured computational output

The full JSON attachment additionally contains every orbit row, every phase/spectrum pair row, and the individual quadrature results. This printed output retains all sixteen named check records and their declared tolerances.

```
{
  "status": "PASS",
  "capture_time_utc": "2026-09-11T04:17:39.474018+00:00",
  "python": "3.12.14",
  "numpy": "2.3.5",
  "scipy": "1.17.0",
  "scope": "Exhaustive stated finite domains and numerical mathematical-model checks; no
↪ observational dataset.",
  "checks": [
    {
      "name": "addition_orbit_period",
      "cases": 19900,
      "status": "PASS",
      "domain": "1<=n<200; 0<=k<n",
      "failures": 0
    },
    {
      "name": "prime_full_orbit_equivalence",
      "cases": 198,
      "status": "PASS",
      "domain": "2<=n<200",
      "failures": 0
    },
    {
      "name": "subgroup_divisor_correspondence",
      "cases": 99,
      "status": "PASS",
      "domain": "1<=n<100",
      "failures": 0
    },
    {
      "name": "base_relabeling_preserves_period",
      "cases": 7076,
      "status": "PASS",
      "bases": [
        2,
        10,
        12,
        60
      ],
      "failures": 0
    },
    {
      "name": "phase_and_spectral_counts",
      "cases": 14400,
      "status": "PASS",
      "domain": "1<=n,m<=120",
      "arithmetic": "fractions.Fraction",
      "failures": 0,
      "example_12_8": [
        12,
        8,
```

```

    4,
    4,
    3,
    6
  ]
},
{
  "name": "canonical_address_loading",
  "cases": 1980,
  "status": "PASS",
  "failures": 0
},
{
  "name": "first_simultaneous_return",
  "cases": 1600,
  "status": "PASS",
  "failures": 0
},
{
  "name": "character_orthonormality",
  "cases": 3,
  "status": "PASS",
  "tolerance": 1e-12,
  "errors": [
    [
      7,
      8.54594767076815e-16
    ],
    [
      12,
      1.866258215684302e-15
    ],
    [
      30,
      3.1267017697858008e-15
    ]
  ]
},
{
  "name": "laplacian_character_identity",
  "cases": 32,
  "status": "PASS",
  "tolerance": 1e-10,
  "max_error": 6.9651443924922745e-15,
  "low_mode_dispersion_ratio": 0.9967913640449618
},
{
  "name": "hermitian_circulant_diagonalization",
  "cases": 1,
  "status": "PASS",
  "max_off_diagonal": 2.2325098619757722e-14,
  "tolerance": 1e-10,
  "seed": 20260911
},
{
  "name": "unitary_norm_and_energy",
  "cases": 400,
  "status": "PASS",
  "max_norm_error": 6.661338147750939e-16,

```

```

    "max_energy_error": 1.3322676295501878e-15,
    "max_unitarity_error": 7.771561172376096e-16,
    "norm_energy_tolerance": 1e-12,
    "unitarity_tolerance": 1e-11
  },
  {
    "name": "energy_shift_and_scale_covariance",
    "cases": 2,
    "status": "PASS",
    "density_shift_error": 1.0591761858187546e-16,
    "density_scale_error": 0.0,
    "tolerance": 1e-11
  },
  {
    "name": "single_mode_density_stationarity",
    "cases": 1,
    "status": "PASS",
    "density_error": 1.1636878843346594e-16,
    "tolerance": 1e-12
  },
  {
    "name": "off_diagonal_perturbation",
    "cases": 4,
    "status": "PASS",
    "values": [
      {
        "epsilon": 0.01,
        "shared_slopes": [
          -1.00054893397624,
          0.9994489284927112
        ],
        "level2_second_order": -0.4099782669531038,
        "level3_second_order": 0.5199788153120721
      },
      {
        "epsilon": 0.003,
        "shared_slopes": [
          -1.0001649038863858,
          0.9998349037383633
        ],
        "level2_second_order": -0.40999804376242976,
        "level3_second_order": 0.5199980931773022
      },
      {
        "epsilon": 0.001,
        "shared_slopes": [
          -1.0000549893149329,
          0.9999449893100639
        ],
        "level2_second_order": -0.4099997814588363,
        "level3_second_order": 0.519999787229608
      },
      {
        "epsilon": 0.0003,
        "shared_slopes": [
          -1.0000164990379552,
          0.9999834990382945
        ],
        "level2_second_order": -0.4099999845803066,

```

```
    "level3_second_order": 0.5199999887474229
  }
],
"shared_slope_tolerance": 0.0001,
"second_order_coefficient_tolerance": 1e-05
},
{
  "name": "lorentz_interval_and_inverse",
  "cases": 2057,
  "status": "PASS",
  "arithmetic": "fractions.Fraction",
  "failures": 0
},
{
  "name": "finite_band_spectrum",
  "cases": 81,
  "status": "PASS",
  "tolerance": 1e-09,
  "max_relative_error": 2.4931664259187294e-14
}
]
}
```